

● PORTFOLIO PROJECT · DATA ANALYTICS

BUSINESS PROBLEM STATEMENT

Stock Performance Analysis

An End-to-End Data Analytics Project covering data collection, database engineering, KPI analysis, and interactive business intelligence reporting.

PREPARED BY

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PROJECT TYPE

Portfolio Project

PYTHON

MYSQL

SQL

POWER BI

GIT

GITHUB

STOCK PERFORMANCE DASHBOARD

Confidential | Prepared for Portfolio Review

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SECTION 01

Executive Summary

Introduction

The stock market generates vast amounts of financial data every trading day. Investors and analysts require accurate, timely, and well-structured information to evaluate company performance and make informed investment decisions. However, raw stock market data alone provides limited business value unless it is cleaned, analyzed, and transformed into meaningful insights.

This project presents an end-to-end Stock Performance Analysis solution that automates the collection of historical stock market data, stores it in a relational database, performs analytical calculations, and visualizes the results through an interactive Power BI dashboard.

The solution demonstrates a complete analytics workflow, from data acquisition to business reporting, using industry-standard tools including **Python, MySQL, SQL, and Power BI**.

5COMPANIES
ANALYZED**6**

PROCESS PHASES

1 YRHISTORICAL DATA
WINDOW**1**INTERACTIVE
DASHBOARD



SECTION 02

Business Problem

Organizations, financial analysts, and retail investors often face challenges when working with historical stock market data because:

- Stock data is distributed across multiple platforms.
- Manual analysis is time-consuming.
- Performance comparisons between companies are difficult without automation.
- Investment decisions require historical trends rather than isolated daily prices.
- Business users require dashboards rather than raw datasets.

Without a structured analytics process, stakeholders spend considerable time collecting, cleaning, and analyzing data before meaningful insights can be generated.



SECTION 03

Project Objectives

The primary objectives of this project are:

- Automate the collection of historical stock market data.
- Store structured financial data in a MySQL database.
- Validate and clean the imported dataset.
- Calculate key performance indicators (KPIs).
- Compare the performance of multiple companies.
- Analyze stock volatility and trading activity.
- Build an interactive Power BI dashboard.
- Demonstrate an end-to-end analytics pipeline.



SECTION 04

Scope of the Project

The project focuses on historical stock performance analysis for five major technology companies over a one-year period.

Company	Stock Symbol
Apple	AAPL
Microsoft	MSFT
Alphabet (Google)	GOOGL
Amazon	AMZN
Tesla	TSLA

Analysis Coverage

- Historical closing prices
- Daily trading volume
- Moving averages
- Total returns & average daily returns
- Stock volatility
- Comparative performance analysis



SECTION 05

Data Source

The dataset was collected using the Yahoo Finance API through the Python **yfinance** library.

Dataset Characteristics

- Historical daily stock prices
- Open price
- High price
- Low price
- Closing price
- Trading volume & trading date

The extracted data was subsequently stored in a MySQL database for further analysis.



SECTION 06

Methodology

The project followed a structured analytics workflow consisting of six phases, moving from raw data acquisition through to business-ready visualization.

1

ACQUISITION

Data Collection

Historical stock data was downloaded using Python from Yahoo Finance.

2

STORAGE

Data Storage

The extracted dataset was loaded into a MySQL database using SQLAlchemy.

3

QUALITY CHECK

Data Validation

Validation scripts were executed to identify missing values, duplicate records, record counts, date ranges, and overall data consistency.

4

ANALYTICS

KPI Analysis

Python scripts calculated Total Return (%), Average Closing Price, Average Daily Return, Average Trading Volume, and Volatility Percentage.

5

PREPARATION

Reporting Dataset

Summary datasets were generated for dashboard reporting.

6

VISUALIZATION

Dashboard Development

Interactive dashboards were created in Microsoft Power BI to visualize business insights.



SECTION 07

Technology Stack

Technology	Purpose
Python	ETL & Data Processing
Pandas	Data Cleaning & Analysis
yFinance	Data Extraction
MySQL	Database Storage
SQLAlchemy	Database Connectivity
SQL	Data Management
Power BI	Dashboard Development
Git	Version Control
GitHub	Project Repository



SECTION 08

Expected Business Benefits

The completed solution provides several business advantages:

- Automated stock data collection
- Faster analytical reporting
- Centralized data storage
- Improved data quality through validation
- Simplified comparison of multiple stocks
- Interactive dashboard for business users
- Improved decision-making through KPI visualization



SECTION 09

Project Deliverables

The project delivers:

- Python ETL scripts
- MySQL database
- SQL queries
- Data validation scripts
- KPI analysis scripts
- Automated master pipeline
- Summary datasets
- Power BI dashboard
- GitHub repository
- Project documentation



SECTION 10

Conclusion

The Stock Performance Analysis Dashboard successfully demonstrates an end-to-end data analytics solution by integrating data collection, database management, analytical processing, and business visualization into a single workflow.

The project highlights practical applications of Python, SQL, MySQL, and Power BI while showcasing industry-standard analytics practices such as ETL development, data validation, KPI reporting, and dashboard design.

This solution serves as a strong foundation for future enhancements, including direct database connectivity with Power BI, automated scheduling, cloud deployment, and advanced financial analytics.

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